

Optimizing Decisions for Personal and Business Development

IEDA 1250

The Hong Kong University of Science and Technology



- 1 Introduction to Queuing Theory
- 2 Little's Law
- 3 M/M/1 Queue Performance Metrics
- 4 Birth and Death Processes

Core Parameters: Rates and Times

Arrivals vs. Service

- **Arrivals: Poisson process**
 - **Average Arrival Rate (λ):** e.g., 3 jobs/sec.
 - **Mean Interarrival Time ($1/\lambda$):** The average time between successive arrivals. (e.g., 1/3 sec).
- **Service: Exponential**
 - **Average Service Rate (μ):** e.g., 4 jobs/sec.
 - **Mean Service Time/ Mean Job Size ($E[S] = 1/\mu$):** The expected time required to service a job. (e.g., $E[S] = 1/4$ sec).

Performance Metrics

- **Time & Counting:**

- T : Response Time/Time in System.
- T_Q : Waiting Time. **Equation:** $E[T] = E[T_Q] + E[S]$.
- N : Number of Jobs in the System (queue + ones being served).
- N_Q : Number of Jobs in Queue.

- **System State:**

- **Throughput** (X_i): The rate of job completions. ($X_i = C/\tau$, where C is completed jobs over time τ).
- **Utilization** (ρ_i): The expected fraction of time the server is busy. ($\rho_i = B/\tau$, where B is busy time over period τ).

Remarks on the Metrics

Stability

- **Stability Rule:** $\lambda < \mu$ for a single server queue.
- To ensure that the queue will not grow indefinitely.

The Throughput Paradox

- Does a faster server (higher μ) increase throughput?
- **No!** As long as the system is stable, the throughput is bottlenecked by arrivals ($X = \lambda$).

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Little's Law

Universal Relationship

Little's Law establishes a fundamental relationship for any stable, open system. It states that the long-term average number of jobs in the system ($E[N]$) equals the average arrival rate (λ) multiplied by the average time a job spends in the system ($E[T]$).

$$E[N] = \lambda \times E[T]$$

Example: Suppose you run a local coffee shop.

- You observe that, on average, **10 customers enter the shop per hour** ($\lambda = 10$ customers/hour).
- Through statistics, you find that each customer spends an average of **0.5 hours** in the shop ($E[T] = 0.5$ hours).

By Little's Law ($E[N] = \lambda \times E[T]$):

$$E[N] = 10 \times 0.5 = 5 \text{ customers}$$

Applying Little's Law

The Concept of a “System”

The true power of Little's Law lies in its flexibility. We can redefine the “system” to derive different performance metrics.

- **Scenario A: The Entire Facility (Queue + Servers)**

- Average jobs in system: $E[N]$
- Average time in system: $E[T]$
- Equation: $E[N] = \lambda \times E[T]$

- **Scenario B: The Waiting Line Only**

- Average jobs in queue: $E[N_Q]$
- Average waiting time: $E[T_Q]$
- Equation: $E[N_Q] = \lambda \times E[T_Q]$

Applying Little's Law: Deriving Utilization

Scenario C: The Server Only

Define our “system” as strictly the server itself, excluding the waiting line.

- **Time spent:** The average time a job spends in the server is its average service time, $E[S]$.
- **The expected number of jobs inside the server:** (The server is either busy processing 1 job or idle with 0 jobs.)

$$E[N_S] = E[\text{\# of jobs in sys.}] = 1 \times P(1 \text{ job in sys.}) + 0 \times P(0 \text{ jobs}) = 1 \times P(\text{device is busy}) = \rho$$

- **Applying the Law** ($E[N] = \lambda \times E[T]$):
 - Substitute the variables: $\rho = \lambda \times E[S]$
 - Since service rate $\mu = 1/E[S]$, we arrive at the **Utilization Law**:

$$\rho = \frac{\lambda}{\mu}$$

Practice: Little's Law

Midtown Bank has **two tellers** on duty. Customers arrive at a mean rate of **40 per hour**. Each teller requires an average of **2 minutes** to serve a customer. Arriving customers wait in a single line for an average of **1 minute** before service begins.

- **Mean Time in System ($E[T]$):**

$$E[T] = E[T_Q] + E[S] = 1 + 2 = \mathbf{3 \text{ minutes}}$$

- **Mean Number in Queue ($E[N_Q]$):**

$$E[N_Q] = \lambda E[T_Q] = \frac{2}{3} \times 1 = \frac{\mathbf{2}}{\mathbf{3}} \text{ customers}$$

- **Mean Number in System ($E[N]$):**

$$E[N] = \lambda E[T] = \frac{2}{3} \times 1 \times 3 = \mathbf{2 \text{ customers}}$$

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Summary of M/M/1 Performance Metrics

1. Expected Number of Jobs (System = Queue + Server)

- **In System:** $E[N] = \frac{\rho}{1-\rho}$
- **In Queue:** $E[N_Q] = \frac{\rho^2}{1-\rho}$
- **In Server (Utilization):** $\rho = \frac{\lambda}{\mu}$

Relationship: $E[N] = E[N_Q] + \rho$

2. Expected Time Spent (System = Queue + Server)

- **Time in System:** $E[T] = \frac{1}{\mu-\lambda}$
- **Waiting Time in Queue:** $E[T_Q] = \frac{\rho}{\mu(1-\rho)}$
- **Service Time:** $E[S] = \frac{1}{\mu}$

Relationship: $E[T] = E[T_Q] + E[S]$ **By Little's Law:** $E[N] = \lambda E[T]$ and $E[N_Q] = \lambda E[T_Q]$

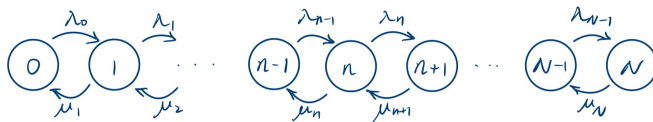
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Birth and Death Processes

Definition

A **Birth and Death Process** is a special type of continuous-time Markov chain (CTMC) with states $0, 1, \dots$ where transitions from state i can only go to adjacent states: either $i-1$ or $i+1$.

- **System State:** Usually represents the size of a population (e.g., the number of jobs in a queueing system).
- **State Transitions:**
 - **Birth (+1):** Occurs when the state increases by 1. The rate of birth is denoted as λ_i .
 - **Death (-1):** Occurs when the state decreases by 1. The rate of death is denoted as μ_i .
- **Exponential Timing & Independence:**
 - When there are i jobs in the system, the time until the next birth $\sim \exp(\lambda_i)$.
 - It is strictly **independent** of the time until the next death $\sim \exp(\mu_i)$.



- M/M/1 and M/M/K are special cases of the Birth and Death Process.
- state n : n jobs in the system
- π_n denotes the probability that the system has i jobs *or* the probability that the process is in state n .

Birth and Death Processes: Balance Equations

Equating the rate the process leaves a state to the rate it enters:

$$\text{State 0: } \lambda_0\pi_0 = \mu_1\pi_1$$

$$\text{State } n > 0: (\lambda_n + \mu_n)\pi_n = \mu_{n+1}\pi_{n+1} + \lambda_{n-1}\pi_{n-1}$$

Rewriting these equations gives:

$$\lambda_0\pi_0 = \mu_1\pi_1$$

$$\lambda_n\pi_n = \mu_{n+1}\pi_{n+1} + (\lambda_{n-1}\pi_{n-1} - \mu_n\pi_n), \quad n \geq 1$$

Through substitution, this equivalently simplifies to:

$$\lambda_0\pi_0 = \mu_1\pi_1$$

$$\lambda_1\pi_1 = \mu_2\pi_2 + (\lambda_0\pi_0 - \mu_1\pi_1) = \mu_2\pi_2$$

$$\lambda_2\pi_2 = \mu_3\pi_3 + (\lambda_1\pi_1 - \mu_2\pi_2) = \mu_3\pi_3$$

$$\vdots$$

Solving Steady-State Probabilities

Solving iteratively in terms of π_0 yields:

$$\pi_1 = \frac{\lambda_0}{\mu_1} \pi_0$$

$$\pi_2 = \frac{\lambda_1}{\mu_2} \pi_1 = \frac{\lambda_1 \lambda_0}{\mu_2 \mu_1} \pi_0$$

$$\pi_3 = \frac{\lambda_2}{\mu_3} \pi_2 = \frac{\lambda_2 \lambda_1 \lambda_0}{\mu_3 \mu_2 \mu_1} \pi_0$$

$\vdots$

$$\pi_n = \frac{\lambda_{n-1}}{\mu_n} \pi_{n-1} = \frac{\lambda_{n-1} \lambda_{n-2} \cdots \lambda_0}{\mu_n \mu_{n-1} \cdots \mu_1} \pi_0$$

Using the normalization condition $\sum_{n=0}^{\infty} \pi_n = 1$, we obtain:

$$1 = \pi_0 + \pi_0 \sum_{n=1}^{\infty} \frac{\lambda_{n-1} \cdots \lambda_0}{\mu_n \cdots \mu_1}$$

Deriving Steady-State Probabilities (π_i) for M/M/1

From the balance equations of the M/M/1 Birth-Death process, we have:

$$\pi_1 = \frac{\lambda}{\mu} \pi_0$$

$$\vdots$$

$$\pi_i = \left(\frac{\lambda}{\mu}\right)^i \pi_0 = \rho^i \pi_0 \quad (\text{where } \rho = \frac{\lambda}{\mu})$$

To find π_0 , we use the normalization condition (probabilities sum to 1):

$$\sum_{i=0}^{\infty} \pi_i = 1 \implies \sum_{i=0}^{\infty} \rho^i \pi_0 = 1$$

$$\implies \pi_0 \left(\frac{1}{1-\rho} \right) = 1 \quad (\text{assuming } \rho < 1 \text{ for stability})$$

$$\implies \pi_0 = 1 - \rho$$

Deriving the Expected Number of Jobs ($E[N]$)

By definition, the expected number of jobs is calculated by summing the product of each possible state i (i jobs in the system) and its corresponding state probability π_i :

$$\begin{aligned} E[N] &= \sum_{i=0}^{\infty} i\pi_i \\ &= \sum_{i=1}^{\infty} i\rho^i(1-\rho) \\ &= \rho(1-\rho) \sum_{i=1}^{\infty} i\rho^{i-1} \\ &= \frac{\rho}{1-\rho} \end{aligned}$$

Practice: Birth-and-Death Process

Problem

Consider a birth-and-death process with the following rates:

- Death rates: $\mu_n = 2$ for all $n = 1, 2, \dots$
 - Birth rates: $\lambda_0 = 3, \lambda_1 = 2, \lambda_2 = 1$
 - Termination: $\lambda_n = 0$ for $n = 3, 4, \dots$
- Display the CTMC diagram.
 - Calculate the steady-state probabilities $\pi_0, \pi_1, \pi_2, \pi_3$, and π_n for $n = 4, 5, \dots$
 - Suppose we define the **effective arrival rate** as:

$$\lambda_e = \sum_{n=0}^{\infty} \lambda_n \pi_n$$

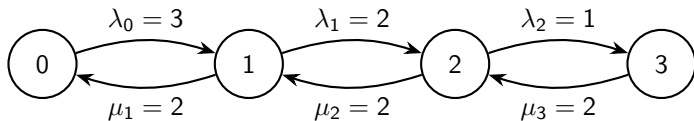
Calculate $E[N]$, $E[N_Q]$, $E[T]$, and $E[T_Q]$ by replacing the arrival rate λ in Little's Law with the effective arrival rate λ_e .

Practice Solution: CTMC Diagram and Balance

Given

$$\lambda_0 = 3, \quad \lambda_1 = 2, \quad \lambda_2 = 1, \quad \lambda_n = 0 \ (n \geq 3),$$
$$\mu_n = 2, \quad n = 1, 2, \dots$$

Only states 0, 1, 2, 3 can have positive steady-state probability.



Local balance equations:

$$3\pi_0 = 2\pi_1, \quad 2\pi_1 = 2\pi_2, \quad \pi_2 = 2\pi_3.$$

Practice Solution: Steady-State Probabilities

From local balance,

$$\pi_1 = \frac{3}{2}\pi_0, \quad \pi_2 = \pi_1 = \frac{3}{2}\pi_0, \quad \pi_3 = \frac{1}{2}\pi_2 = \frac{3}{4}\pi_0.$$

Use normalization:

$$\pi_0 + \pi_1 + \pi_2 + \pi_3 = 1.$$

Substitute the expressions in terms of π_0 :

$$\pi_0 + \frac{3}{2}\pi_0 + \frac{3}{2}\pi_0 + \frac{3}{4}\pi_0 = 1.$$

$$\frac{19}{4}\pi_0 = 1 \quad \Rightarrow \quad \pi_0 = \frac{4}{19}.$$

Therefore,

$$\pi_0 = \frac{4}{19}, \quad \pi_1 = \frac{6}{19}, \quad \pi_2 = \frac{6}{19}, \quad \pi_3 = \frac{3}{19}$$

and

$$\pi_n = 0, \quad n = 4, 5, \dots$$

Practice Solution: Effective Arrival Rate and Queue Size

The effective arrival rate is

$$\lambda_e = \sum_{n=0}^{\infty} \lambda_n \pi_n = 3\pi_0 + 2\pi_1 + \pi_2 = \frac{12}{19} + \frac{12}{19} + \frac{6}{19} = \frac{30}{19}.$$

Expected number in the system:

$$E[N] = \sum_{n=0}^3 n\pi_n = \frac{6}{19} + \frac{12}{19} + \frac{9}{19} = \frac{27}{19}.$$

For a single-server system, queue length in state n is $\max\{n-1, 0\}$. Thus,

$$E[N_Q] = (0)\pi_1 + (1)\pi_2 + (2)\pi_3 = \frac{6}{19} + \frac{6}{19} = \frac{12}{19}.$$

Practice Solution: Time Metrics

By Little's Law with λ_e ,

$$E[T] = \frac{E[N]}{\lambda_e} = \frac{27/19}{30/19} = 0.9$$

$$E[T_Q] = \frac{E[N_Q]}{\lambda_e} = \frac{12/19}{30/19} = 0.4$$

Final answers:

$$\pi_0 = \frac{4}{19}, \quad \pi_1 = \frac{6}{19}, \quad \pi_2 = \frac{6}{19}, \quad \pi_3 = \frac{3}{19}, \quad \pi_n = 0 \quad (n \geq 4)$$

$$\lambda_e = \frac{30}{19}, \quad E[N] = \frac{27}{19}, \quad E[N_Q] = \frac{12}{19}, \quad E[T] = 0.9, \quad E[T_Q] = 0.4$$

Solving Birth-and-Death Process Problems

① Define States & Draw CTMC Diagram

- Identify all the states. Look for termination rates (e.g., $\lambda_n = 0$) to determine if the system is finite or infinite.
- Identify the transition rates between states (e.g., λ_i, μ_i).

② Establish Balance Equations

- For any adjacent states, the rate of leaving must equal the rate of entering.
- Express every π_i as a function of π_0 (e.g., $\pi_1 = \frac{\lambda_0}{\mu_1} \pi_0$).

③ Apply the Normalization Condition

- Use $\sum_{n=0}^{\infty} \pi_n = 1$ to solve for π_0 .
- Once π_0 is found, determine the values for all other steady-state probabilities π_n .

④ Derive Performance Metrics

- **System Size:** $E[N] = \sum n\pi_n$.
- **Little's Law:** Use the *effective* rate λ_e : the "true" flow entering the system.
time metrics: $E[T] = \frac{E[N]}{\lambda_e}$ and $E[T_Q] = \frac{E[N_Q]}{\lambda_e}$.

Solving Birth-and-Death Process Problems

Calculating $E[N_Q]$ in Single-Server Systems In this problem, the death rates are constant: $\mu_1 = \mu_2 = \mu_3 = 2$

- This implies a **single-server** system where at most **one** job can be in service at any time.

Formula for $E[N_Q]$

Substituting this into the expectation formula:

$$E[N_Q] = \sum_{n=1}^K (n-1)\pi_n, \quad K=3 \text{ for this problem.}$$

Note: This $(n-1)$ logic only holds when there is exactly one server. If there were k servers, we would use $(n-k)$ for $n > k$.

Thank you for listening !